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Faculty of Information Technology

Department: All

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Group (s): A, B, E & F

Question Breakdown:

1. *Section A: True or False Questions*
2. *Section B: Multiple Choice Questions (MCQs)*
3. *Section C: Practical*
4. *Section D: Problem Statement*

Section A: True or False Questions

1. Normalization in SQL involves reducing redundancy and organizing data into smaller, related tables to improve data integrity.
2. If a transaction fails after some operations have been completed, atomicity ensures that those operations are rolled back and do not remain in the database.
3. The DISTINCT keyword in SQL is used to remove duplicate values from query results, ensuring unique rows are returned.
4. All tables in an Oracle Database must reside in the same tablespace.
5. You can specify a default tablespace for users in Oracle Database.
6. Business process modeling in Management Information Systems helps organizations visualize, analyze, and improve workflows.
7. Datatype constraints in SQL enhance data integrity, validation, and security by defining rules for data storage and operations.
8. PL/SQL allows the creation of reusable procedures and functions to simplify database operations.
9. An IN parameter in a stored procedure can be modified within the procedure.
10. Using the %ISOPEN attribute for implicit cursors always returns TRUE after executing a SQL statement.

11. In Oracle PL/SQL, the command `ALTER SESSION SET CONTAINER = CDB%ROOT` switches to the root container in a multitenant database architecture.
12. Using `AUTHID CURRENT_USER` ensures that a procedure runs with the caller's privileges instead of the definer's privileges.
13. The `EXCEPTION` block in PL/SQL ensures proper error handling, such as rolling back transactions if an error occurs.
14. Subqueries in SQL can be used to find employees earning above the department's average salary.
15. When calling a stored procedure with an `OUT` parameter, the output value must be assigned to a variable in the calling environment.
16. In SQL, using projection to select more columns increases the number of rows returned by a query.
17. In relational databases, data redundancy is reduced by organizing data into related tables rather than storing all data in one table.
18. SQL triggers can be used to automatically execute certain actions in response to database events such as `INSERT` or `DELETE`.
19. ACID properties in databases ensure data consistency, even in cases of transaction failure or system crashes.
20. The `%ROWTYPE` attribute in PL/SQL allows a variable to inherit the structure of a table row or cursor.
21. SQL allows the creation of custom exceptions for specific error-handling scenarios in stored procedures.
22. When using the `GROUP BY` clause in SQL, all columns in the `SELECT` statement must either be aggregated or included in the `GROUP BY` clause.
23. The `TRUNCATE` command in SQL removes all rows from a table and cannot be rolled back.
24. Implicit cursors in PL/SQL require explicit opening and closing by the programmer.
25. The `COMMIT` command ensures that all changes made in a transaction are permanently saved to the database.

Section A: True or False Questions with answers

Atomicity

1. **True:** Atomicity ensures that a transaction is completed fully or not at all, meaning that all operations within a transaction must succeed for the transaction to be considered successful.
2. **False:** If a transaction fails after some operations have been completed, atomicity ensures that those operations are rolled back and do not remain in the database.
3. **True:** Atomicity is one of the key properties of a relational database management system (RDBMS) that helps maintain data integrity.
4. **False:** Atomicity does not guarantee that once a transaction is committed, it cannot be rolled back. This is a property of durability, not atomicity. Atomicity is about ensuring transactions are all-or-nothing.

5. **False:** The concept of atomicity is closely related to the concepts of consistency and durability in database transactions, as they are all part of the ACID properties.

SQL Projection

6. **True:** Projection in SQL refers to selecting specific columns from a table rather than retrieving all columns.
7. **True:** The SQL SELECT statement is used to perform projection operations.
8. **False:** When using projection, the number of rows returned by a query can be increased by selecting more columns.
9. **True:** Projection can be combined with selection in a single SQL query to filter rows and specify which columns to return.
10. **False:** In SQL, projection can only be performed on numeric data types.
11. **True:** The result of a projection operation can include calculated fields, such as the result of a mathematical operation on selected columns.
12. **True:** Projection does not affect the underlying data in the database; it only changes the output of the query.
13. **True:** You can use the DISTINCT keyword in a projection to eliminate duplicate values from the result set.

PL/SQL Features

14. **True:** PL/SQL provides a secure environment for executing database operations.
15. **False:** PL/SQL does not support code reusability.
16. **True:** PL/SQL includes control structures such as loops and conditionals.
17. **False:** PL/SQL applications are not portable across different operating systems.
18. **True:** PL/SQL allows for the creation of reusable procedures and functions.

Oracle Tablespaces

19. **True:** A tablespace is a logical storage unit in an Oracle Database.
20. **False:** Tablespaces can only contain data files.
21. **True:** The SYSTEM tablespace is automatically created when the database is created.
22. **False:** Undo tablespaces are used to store data dictionary information.
23. **True:** A tablespace can contain multiple data files.
24. **False:** Tablespaces are not necessary for organizing data in an Oracle Database.
25. **True:** You can specify a default tablespace for users in Oracle Database.
26. **True:** Tablespaces can help improve performance by organizing data efficiently.
27. **False:** All tables in an Oracle Database must reside in the same tablespace.
28. **True:** Tablespaces can be used to manage storage allocation for different types of data.

Business Process Modeling (BPM), MIS, and Conceptual Modeling

- 29. **True:** A business process model represents the flow of activities and decisions in an organization.
- 30. **False:** Conceptual modeling focuses solely on the technical details of a system, ignoring business processes.
- 31. **True:** In Management Information Systems, business process modeling helps to visualize and optimize workflows within an organization.
- 32. **False:** A conceptual model in MIS is primarily concerned with the physical structure of the database and network.
- 33. **True:** The goal of business process modeling is to improve efficiency by identifying inefficiencies and potential areas of improvement.
- 34. **False:** In a conceptual model, the focus is to show detailed data processing logic, including algorithms and code.
- 35. **True:** Conceptual models in MIS often use diagrams such as Entity-Relationship (ER) models or UML to represent high-level system structures.
- 36. **False:** Business process modeling only applies to large organizations and is not relevant for smaller businesses.
- 37. **True:** The Business Process Model and Notation (BPMN) is commonly used to standardize the representation of business processes.
- 38. **False:** Conceptual modeling in MIS is not necessary if an organization already has a fully implemented database system.

Oracle Multitenant Architecture

- 39. **True:** A Container Database (CDB) in Oracle can only contain one Pluggable Database (PDB).
- 40. **True:** The root container (CDB\$ROOT) holds system-wide metadata and common objects shared by all Pluggable Databases (PDBs).
- 41. **True:** The PDB\$SEED is a template database used to quickly create new PDBs in Oracle's multitenant architecture.
- 42. **True:** Pluggable Databases (PDBs) can be unplugged from one CDB and plugged into another, enabling portability.
- 43. **True:** In Oracle's multitenant architecture, each Pluggable Database (PDB) shares system resources like CPU and memory with other PDBs within the same CDB.
- 44. **False:** The root container (CDB\$ROOT) stores user data in Oracle databases.
- 45. **True:** Oracle's multitenant architecture helps in simplifying database consolidation by allowing multiple PDBs to exist within a single CDB.
- 46. **True:** When a user connects to a PDB, they can access data and schema objects specific to that PDB, but not data from other PDBs within the same CDB.
- 47. **True:** In Oracle, you can manage multiple databases using a single Container Database (CDB) that consolidates the resources of several Pluggable Databases (PDBs).

48. **True:** Oracle Enterprise Manager (OEM) can be used to monitor and manage both CDBs and PDBs in an Oracle database environment.

Section B: Multiple Choice Questions (MCQs)

1. **What is the main purpose of using indexes in relational databases?**
 - A. To improve query performance by reducing search time.
 - B. To ensure that relationships between tables are preserved.
 - C. To increase the size of the database.
 - D. To manage duplicate records in the database.

2. **What is a potential drawback of using database views?**
 - A. Views cannot filter data dynamically.
 - B. Views consume extra storage in the database.
 - C. Performance may degrade for complex queries executed via views.
 - D. Views cannot be updated under any circumstances.

3. **What role does a pull request serve in Git-based development?**
 - A. To merge changes from one branch into another after peer review.
 - B. To synchronize the local repository with the remote repository.
 - C. To create a backup of the source code.
 - D. To delete outdated branches from the repository.

4. **How does horizontal scaling differ from vertical scaling in databases?**
 - A. Horizontal scaling involves adding resources to a single machine.
 - B. Horizontal scaling spreads the database workload across multiple machines.
 - C. Horizontal scaling increases the memory capacity of a server.
 - D. Horizontal scaling restricts access to concurrent users.

5. **Which of these diagrams is best suited to map a sequence of interactions in a business process?**
 - A. Activity Diagram
 - B. Gantt Chart
 - C. Sequence Diagram
 - D. Entity-Relationship Diagram

6. **Which SQL command is correct for listing all Pluggable Databases (PDBs) in an Oracle system?**
 - A. SHOW PDBs;
 - B. SELECT * FROM DBA_PDBS;
 - C. DISPLAY PDBs;
 - D. LIST ALL PDBs;

7. In the following PL/SQL block, how does the program handle errors?

```
DECLARE

    TYPE user_array IS TABLE OF VARCHAR2(30);

    users user_array := user_array('user1', 'user2', 'user3');

BEGIN

    FOR i IN users.FIRST..users.LAST LOOP

        BEGIN

            EXECUTE IMMEDIATE 'DROP USER ' || users(i) || ' CASCADE';

            DBMS_OUTPUT.PUT_LINE(users(i) || ' has been removed.');
```

EXCEPTION

WHEN OTHERS THEN

DBMS_OUTPUT.PUT_LINE('Error dropping user: ' || SQLERRM);

END;

END LOOP;

END;

- A. The program continues execution for all users even if one fails.
- B. The loop stops processing on the first error.
- C. The program ends completely when an error occurs.
- D. Users are dropped without any error handling.

8. Which function in SQL is commonly used to handle null values?

- A. NULLIFY
- B. NVL
- C. COALESCE
- D. Both B and C

9. Why is version control critical in collaborative software development?

- A. It allows multiple developers to work on the same code without conflicts.
- B. It automatically generates documentation for the project.
- C. It identifies security vulnerabilities in the codebase.
- D. It ensures projects are only accessible offline.

10. **Which of the following is a key feature of transactional systems in RDBMS?**

- A. Transactions must always be rolled back.
- B. They follow the ACID properties to ensure data reliability.
- C. Transactions are executed without any isolation.
- D. Durability is ignored during transaction failures.

11. **What is the primary focus of process modeling in business systems?**

- A. Representing decision-making strategies visually.
- B. Identifying bottlenecks and inefficiencies in workflows.
- C. Documenting end-user responsibilities and roles.
- D. Storing data structures for business operations.

12. **Which of the following is NOT typically included in a high-level conceptual model for a system?**

- A. Relationships between entities
- B. Abstract data elements and processes
- C. Detailed storage and indexing structures
- D. Functional areas of the system

13. **What makes Oracle PDBs ideal for use in multi-tenant environments?**

- A. They allow for better data isolation and resource control.
- B. They integrate seamlessly with NoSQL databases.
- C. They support multiple operating systems in one database.
- D. They eliminate the need for database recovery procedures.

14. **What is the purpose of Git Bash in development workflows?**

- A. To execute Git commands via a command-line interface.
- B. To provide a graphical user interface for managing repositories.
- C. To track changes to files automatically.
- D. To visualize codebase changes in a dashboard.

15. **Which SQL operation would update all null values in a column to a default value?**

- A. Use UPDATE with a condition checking for IS NULL.
 - B. Use the NULLFILL function to replace empty fields.
 - C. Use REPLACE to swap nulls for the default value.
 - D. Use INSERT to overwrite null entries.
-

Additional Questions

16. **What is the role of ACID compliance in databases?**

- A. Ensures data is stored across multiple servers.
 - B. Guarantees reliability and consistency of transactions.
 - C. Prevents any changes to existing records.
 - D. Facilitates real-time processing of large datasets.
-

17. **Which element is essential for designing scalable cloud database systems?**

- A. Centralized architecture for single-point control.
- B. Redundant storage for fault tolerance and reliability.
- C. Fixed server resources for predictable performance.
- D. Complex indexes for every query optimization.

18. **What is the primary function of an audit trigger in Oracle?**

- A. To automatically back up the database after each transaction.
 - B. To track and record changes to specified database objects.
 - C. To delete outdated records from the audit logs.
 - D. To optimize query execution plans.
-

17. **Which of the following is NOT a valid type of Oracle trigger?**

- A. AFTER trigger
 - B. BEFORE trigger
 - C. INSTEAD OF trigger
 - D. ON DELETE trigger
-

18. **In Oracle PL/SQL, which statement is used to declare a cursor for processing a query result set?**

- A. DECLARE CURSOR cursor_name IS query;
 - B. CREATE CURSOR cursor_name AS query;
 - C. OPEN CURSOR cursor_name FOR query;
 - D. FETCH cursor_name INTO variable;
-

19. **What is the purpose of an Oracle package?**

- A. To store static SQL queries and execute them periodically.
 - B. To group related PL/SQL procedures, functions, and variables for modular programming.
 - C. To manage user privileges and roles in the database.
 - D. To enhance the performance of select queries with complex joins.
-

20. **How does an Oracle object type differ from a relational table?**

- A. An object type defines a class with attributes and methods, while a relational table stores data in rows and columns.
- B. An object type is designed for storing only string data, while relational tables store numeric

data.

C. An object type is a special index structure used for fast retrieval, while a relational table is used for data manipulation.

D. There is no difference; object types and relational tables are the same.

21. What does an object table in Oracle allow you to do?

A. It enables you to store and manipulate data using object types, much like a regular relational table, but with additional features for object-oriented programming.

B. It acts as a storage for non-relational data, like JSON or XML.

C. It stores metadata for all objects within the database.

D. It serves as a backup for traditional tables in case of corruption.

22. Which statement is true about an audit trigger in Oracle?

A. Audit triggers cannot be used on system-level objects.

B. Audit triggers are typically used for logging SELECT queries only.

C. Audit triggers can automatically store changes made to database objects in an audit table.

D. Audit triggers only work with DML (Data Manipulation Language) operations, such as INSERT and UPDATE.

23. What is the correct syntax to define a PL/SQL package in Oracle?

A.

```
CREATE PACKAGE package_name AS  
    PROCEDURE procedure_name;  
    FUNCTION function_name RETURN data_type;  
END package_name;
```

B.

```
CREATE PACKAGE package_name FUNCTION function_name;
```

C.

```
DEFINE PACKAGE package_name;
```

D.

```
CREATE PACKAGE package_name PROCEDURE procedure_name;
```

24. Which of the following is a benefit of using a PL/SQL package in Oracle?

- A. Packages allow for the storage of relational data in memory.
 - B. Packages help in reducing network traffic by compiling all necessary procedures into a single unit.
 - C. Packages require frequent manual updates for security.
 - D. Packages prevent users from accessing database objects directly.
-

25. How can a cursor in PL/SQL be used to iterate over query results?

- A. By declaring a cursor and then fetching and processing each record in the result set.
 - B. By using a cursor to store the result set in a global variable for later use.
 - C. By defining a cursor and applying it directly to SELECT statements only.
 - D. Cursors cannot be used to process query results directly.
-

26. What is a cursor variable in PL/SQL?

- A. A pointer that can hold multiple result sets for future manipulation.
 - B. A temporary variable used to store user input.
 - C. A variable that stores the execution plan for a SQL query.
 - D. A variable used to define the type of a database column.
-

27. Which statement about Oracle object types is true?

- A. Object types cannot contain methods or functions.
- B. Object types are useful for representing complex data structures like customer and order information.
- C. Object types are limited to only string-based attributes.
- D. Object types cannot be used with relational tables.

28. Which of the following is the correct syntax to instantiate an object from an object type in Oracle?

- A. `DECLARE`
`obj_name object_type := object_type(attribute1, attribute2);`
 - B. `CREATE obj_name TYPE object_type(attribute1, attribute2);`
 - C. `obj_name := NEW object_type(attribute1, attribute2);`
 - D. `DEFINE obj_name AS object_type(attribute1, attribute2);`
-

29. What type of data does an object table in Oracle store?

- A. It stores data as objects, with attributes defined by object types, which may include complex data structures.
- B. It stores metadata about system-level objects only.
- C. It stores large text and binary data such as images or audio files.
- D. It only stores relational data that is compatible with SQL.

Section C: Practical

1. Trigger-related Questions

a. Understanding Trigger Syntax

Question: What does the following line of code in a trigger signify?

AFTER UPDATE OF salary ON employees

Options:

- a) It specifies that the trigger fires before any updates are made on the salary column of the employees table.
- b) It specifies that the trigger fires after an update is made specifically to the salary column in the employees table.
- c) It triggers an insert operation on the salary column whenever it is updated.
- d) It checks for any changes to the salary column, whether before or after the update.

b. Identifying Trigger Variables

Question: In the trigger code below, what does the :NEW and :OLD keyword represent?

INSERT INTO audit_log (emp_id, old_salary, new_salary, change_date)

VALUES (:OLD.emp_id, :OLD.salary, :NEW.salary, SYSDATE);

Options:

- a) :NEW refers to the value of the column after the update, while :OLD refers to the value before the update.
- b) :NEW refers to the value before the update, while :OLD refers to the value after the update.
- c) :NEW is used for the INSERT statement, and :OLD is used for DELETE operations.
- d) Both :NEW and :OLD refer to the newly inserted values in the table.

2. Procedure-related Questions

a. Identifying Procedure Parameters

Question: What is the purpose of the following parameter in the procedure definition?

PROCEDURE update_total_salary(department_id IN NUMBER) AS

Options:

- a) It defines the procedure's return type.
- b) It defines the procedure's local variable.
- c) It specifies that the department_id value is passed to the procedure when it is executed.
- d) It defines the table from which the data will be updated.

b. Understanding the Logic of a Procedure

Question: In the following procedure, what does the line of code below do?

*UPDATE employees SET salary = salary * 1.05 WHERE department_id = department_id;*

Options:

- a) It updates the salary of all employees, increasing it by 5%.
- b) It updates the salary for employees in a specific department, increasing it by 5%.
- c) It causes an error because the department_id is ambiguous.
- d) It decreases the salary for employees in the specified department by 5%.

c. Procedure Execution

Question: When you execute the following statement, what does it trigger in the database?

EXEC update_total_salary(10);

Options:

- a) It will insert a new record into the employees table with department_id = 10.
- b) It will update the salary for all employees in department 10, increasing their salary by 5%.
- c) It will remove the record of all employees in department 10.
- d) It will calculate the total salary of department 10 but not update it.

3. Function-related Questions

a. Function Output Identification

Question: What will be the output of the following function when called with emp_id = 1?

SELECT calculate_annual_salary(1) FROM dual;

Options:

- a) It will return the monthly salary for employee 1.

- b) It will return the annual salary, i.e., monthly salary * 12, for employee 1.
- c) It will return a NULL value because emp_id does not exist.
- d) It will return the number of years employee 1 has been employed.

b. Analyzing Function Logic

Question: What is the purpose of the following line in the function definition?

*SELECT salary * 12 INTO annual_salary FROM employees WHERE emp_id = emp_id;*

Options:

- a) It calculates the annual salary by multiplying the employee's monthly salary by 12 and stores the result in the variable annual_salary.
 - b) It calculates the total salary for the company by summing up the salary of all employees.
 - c) It retrieves the employee's salary and directly returns the result without any further calculation.
 - d) It generates an error because emp_id is not declared in the function.
-

4. Package-related Questions

a. Identifying Package Components

Question: What is the role of the following line in a package specification?

PROCEDURE insert_employee(p_emp_id IN NUMBER, p_emp_name IN VARCHAR2, p_salary IN NUMBER);

Options:

- a) It defines a function that calculates the employee salary.
- b) It defines a procedure that inserts a new employee into the employees table.
- c) It creates a procedure for deleting employees from the database.
- d) It defines a procedure for updating an employee's salary.

b. Understanding Package Body Logic

Question: In the package body code below, what does the COMMIT; statement inside the insert_employee procedure signify?

PROCEDURE insert_employee(p_emp_id IN NUMBER, p_emp_name IN VARCHAR2, p_salary IN NUMBER) IS
BEGIN

```
INSERT INTO employees (emp_id, emp_name, salary)
VALUES (p_emp_id, p_emp_name, p_salary);
COMMIT;
END insert_employee;
```

Options:

- a) It ensures that any changes made to the employees table are saved to the database.
- b) It triggers a rollback if the insertion fails.
- c) It creates a temporary backup of the employees table.
- d) It forces an automatic commit after every procedure execution.

c. Calling Package Procedures

Question: When you execute the following statement, what happens in the database?

```
EXEC employee_pkg.insert_employee(6, 'Alice Cooper', 6000);
```

Options:

- a) A new employee with ID 6, name 'Alice Cooper', and salary 6000 will be inserted into the employees table.
 - b) The employee with ID 6 will be updated with name 'Alice Cooper' and salary 6000.
 - c) A new record will be added to the audit_log table with information about employee Alice Cooper.
 - d) It will generate an error because the employee data already exists in the table.
-

5. General PL/SQL Questions

a. Identifying Error Handling

Question: What happens if an error occurs inside the following PL/SQL block?

```
BEGIN
    UPDATE employees SET salary = salary * 1.05;
EXCEPTION
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE('An error occurred');
END;
```

Options:

- a) The error will be caught and a message saying "An error occurred" will be displayed.
- b) The error will cause the entire procedure to fail and no message will be displayed.
- c) The update will be rolled back automatically without displaying any message.
- d) The update will still be executed, and the error message will be ignored.

b. Understanding Return Statements

Question: What does the following return statement in a function do?

RETURN annual_salary;

Options:

- a) It sends the value of annual_salary back to the calling environment, ending the function execution.
- b) It updates the annual_salary column in the employees table.
- c) It prints the value of annual_salary to the screen.
- d) It stops the execution of the trigger and proceeds with other operations.

SECTION D. Problem Statement

You are tasked with managing a database system for an e-commerce platform. The system needs to handle orders, customers, and payments effectively. Below is a detailed breakdown of the tasks that need to be implemented:

1. Procedure with Cursor:

- Write a PL/SQL procedure that retrieves all pending orders from the orders table and updates their status to "Processed". The procedure should use a cursor to iterate over all pending orders, fetch their details (e.g., OrderID, CustomerID, OrderAmount), and process them.

2. Function:

- Create a PL/SQL function that calculates the discount for a customer based on their total spending. The function should accept CustomerID as an input and return the discount percentage as a numeric value. The discount calculation should be based on the following conditions:
 - If the total amount spent by the customer is greater than \$1000, they receive a 10% discount.
 - If the total amount spent is between \$500 and \$1000, they receive a 5% discount.

- Otherwise, they receive no discount.

3. **Package:**

- Write a PL/SQL package that includes both the procedure and function from the above tasks. The package should be structured with the procedure process_orders and the function calculate_discount. The package should also have a specification and body, where the procedure and function are encapsulated within it.

4. **Trigger:**

- Develop a trigger that automatically updates the last_updated field of the customers table whenever a new order is placed. The trigger should fire after an insert operation on the orders table and update the corresponding customer's last_updated date in the customers table.

Guidelines:

- Ensure that the code is clear, well-organized, and follows PL/SQL best practices.
- Test your procedure and function to ensure they perform correctly when invoked.
- The package should correctly encapsulate the procedure and function, making it reusable.
- The trigger should automatically update the customer's information when an order is placed, without manual intervention.

Objective:

By completing this task, you will demonstrate your understanding of how to combine procedures, functions, cursors, packages, and triggers in a real-world scenario. This will also test your ability to design, implement, and maintain PL/SQL components in a database system.

Final Thoughts:

As you prepare for this exam, remember that understanding the underlying logic is the key to success. The concepts, structures, and processes that we have discussed throughout the course are what truly matter. I strongly encourage you to engage in discussions with your peers and collaborate within your groups. The more you understand the logic behind the topics, the more confident you will be in tackling any question. I believe that if you grasp the logical foundations of the material, you will be well on your way to achieving a grade of 80% or higher.

However, do not solely rely on this task. It is important to read your notes thoroughly, review past topics, and explore additional resources available on the internet. This will deepen your understanding and help you approach the exam with a well-rounded perspective.

Good luck with your preparation, and I am confident you will succeed!

Blessings to you all!